Smartcampus

A Digital Platform for Optimizing Student Services

IS 436 - IS436_1143_FA2025

Deliverable 5 / Presentation

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NextGen Solutions:

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Overview of the Project: Introduction, Timeline, and Stakeholders

Introduction:

- Sponsor: Shiva Sharma, UMBC
- Purpose: Create a centralized digital platform for advising, tutoring, SI sessions, and IT support
- Problem Solved:
  - Services currently spread across multiple websites
  - Inconsistent booking and notifications
  - No single dashboard for students

Timeline:

- Total Project Length: 12 weeks
- Methodology: Agile Scrum (5 Sprints)

Overview of the Project: Introduction, Timeline, and Stakeholders

Stakeholders

- Project Sponsor: Shiva Sharma
- Primary Users: UMBC Students
- Secondary Users: Advisors, Tutors, SI Leaders
- Admin Users: Department & System Administrators
- Development Team: NextGen Solutions

Financial Projection: Estimated Project Costs

Cost Categories:

- Software Development (custom system)
- Cloud Hosting & Server Resources
- Database & Backup Storage
- Testing & QA Tools
- Maintenance & Support

Estimated Financial Summary:

- Initial Development Cost: High upfront (custom development)
- Operating Cost: Moderate (cloud + maintenance)
- Break-even Point: Year 2
- Total Net Benefit by Year 3: $60,000

Year	Development Cost	Maintenance	Staff Time Savings	Admin Cost Savings	Net Benefit
2024	150,000 - system development, vendor integration, training, implementation	20,000	25,000 - early adoption by students, reduced walk-in scheduling	10,000 - streamlined admin reporting and communication	-135,000
2025	40,000 - updates and integrations	25,000	80,000 - staff spend less time scheduling, more automated	40,000 - reduced duplication across all departments, reporting tools	+55,000
2026	15,000 - routine updates and support	25,000	120,000- staff workloads reduced significantly	60,000 - reduced overhead from unified platform	+140,000

Interview Insights: Key Questions Explored During Interviews

Sponsor Interview

- What are the top problems students report when seeking services?
- How do staff currently manage schedulting?
- Which university systems must we integrate with?
- What metrics define success?
- Are there policy constraints on data retention?

Student Interview

- Can you describe a time when you or other students struggled to use a booking system for academic appointments or events?
- We asked students to rank factors that contribute to confusion when booking.
- We asked students to rank their preferred methods for receiving booking confirmations or reminders
- Would you like to receive SMS reminders? If so, what type of information would you want included in those messages?

Advisor Interview

- How do advisors manage availability?
- What issues exist with missed appointments?

Interview Insights: Key Questions Explored During Interviews

Insights from Interviews:

- Biggest challenges is system fragmentation/need for real-time tracking and analytics
- Need for SSO integration
- Success = 20% increase in advising visits
- 30% faster ticket resolution
- Students miss appointments due to confusing systems
- Strong preference for SMS reminders
- Calendar synchronization is critical
- No-shows are a serious workload issue

User Needs: Essential User Requirements

User Requirement 1

- Students need a single dashboard to:
  - View services
  - Book appointments
  - Track tickets

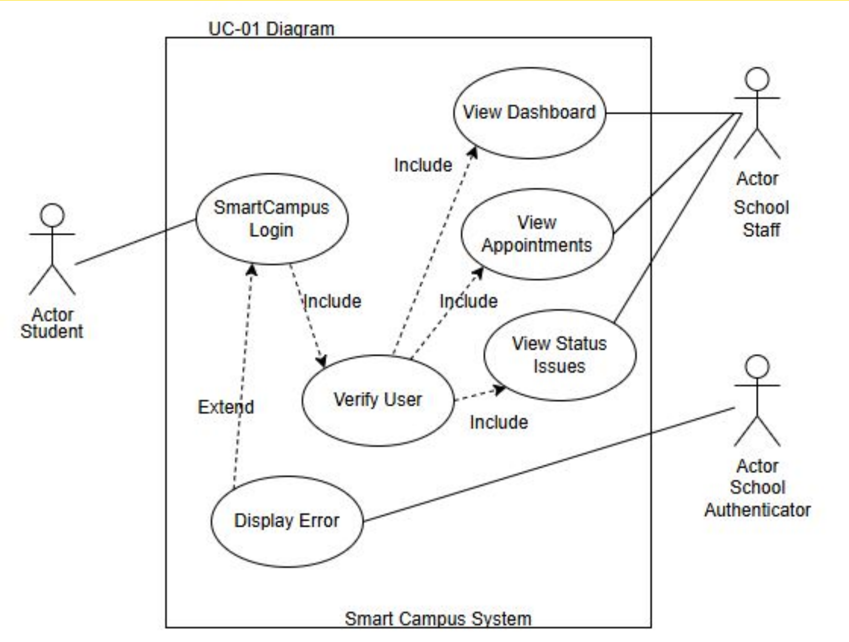
User Requirement 2

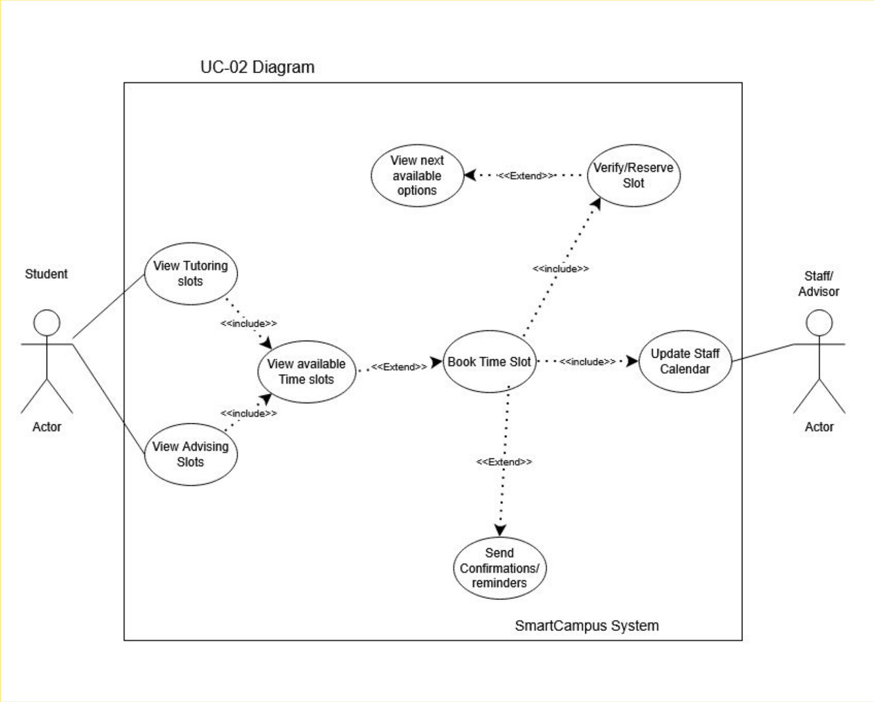
- Students and staff need real-time scheduling with:
  - Live availability
  - Automated reminders
  - Calendar synchronization

User Requirement 3

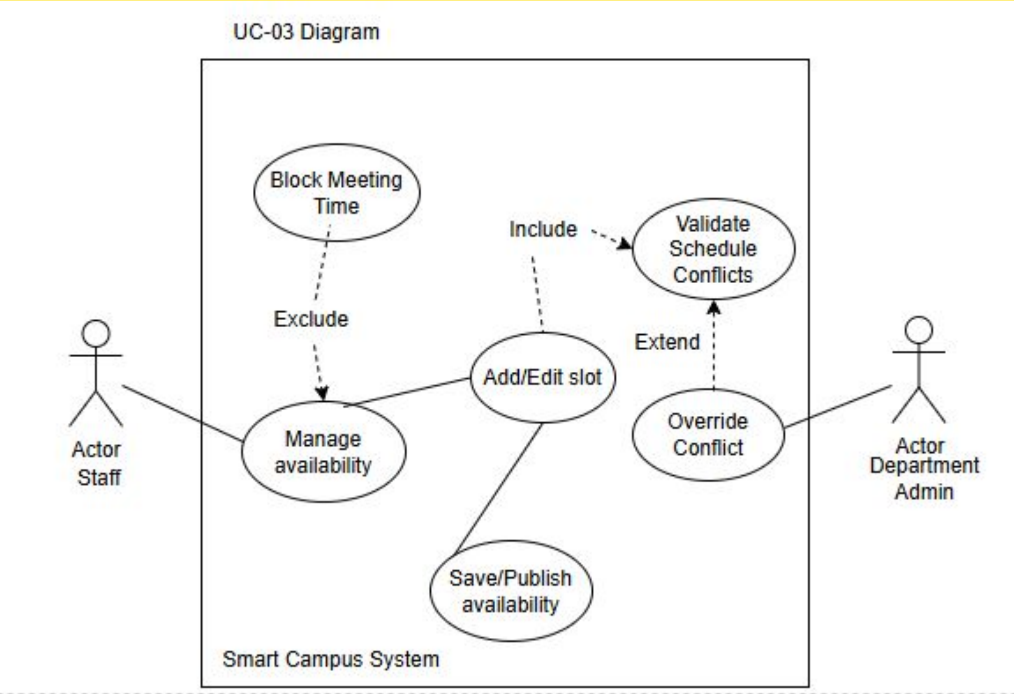
- Admins need:
  - Reporting dashboards
  - Usage metrics
  - Performance tracking

Visualizing Interaction: Use Case Diagrams 1 and 2

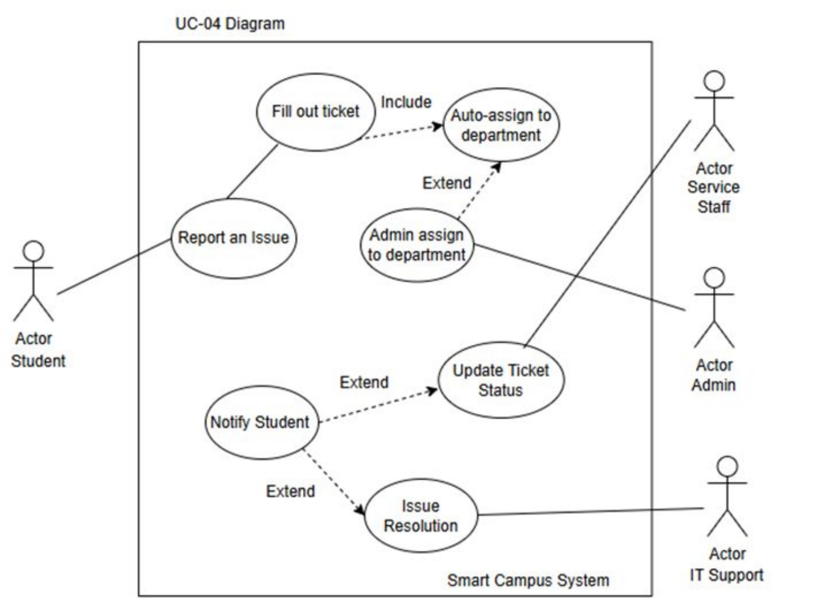


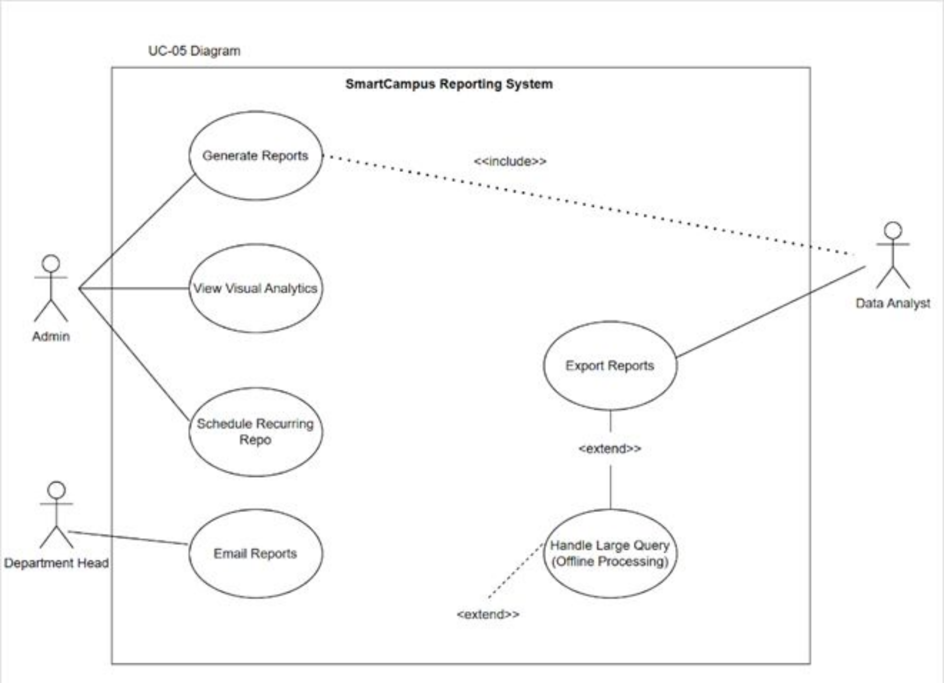


Visualizing Interaction: Use Case Diagrams 3



Visualizing Interaction: Use Case Diagrams 4 and 5





Database Design: Entity-Relationship Diagram (ERD) with database script.

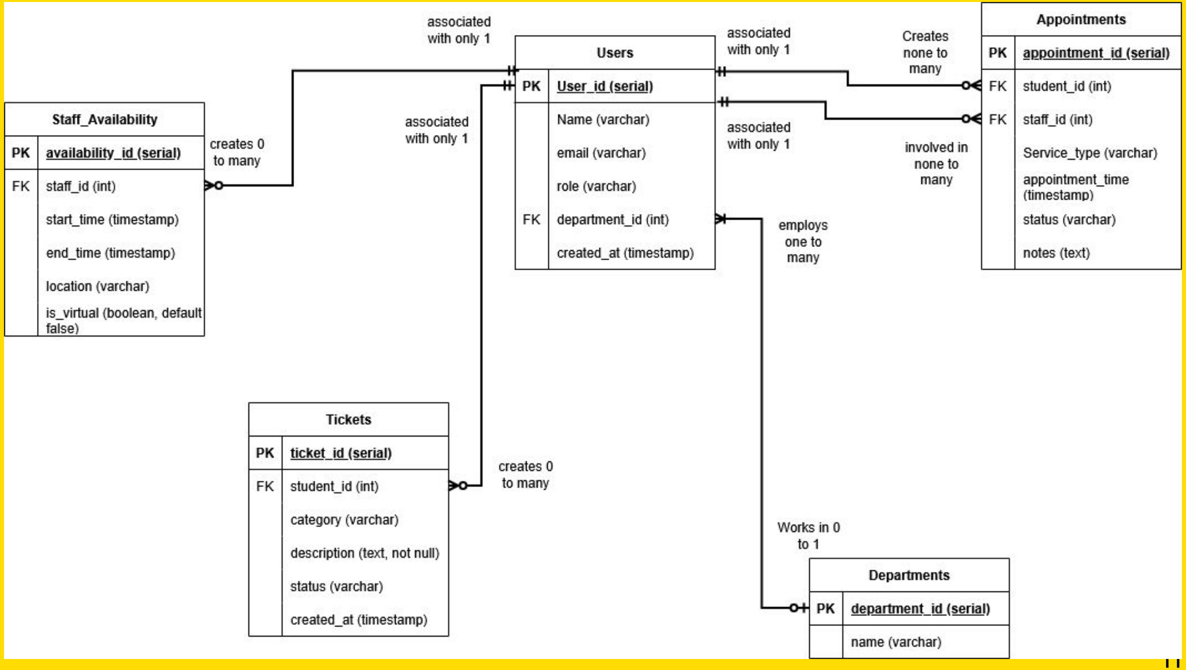


Table creation:
https://github.com/wbt6/docker-database-project/blob/main/init-db/01_create_tables.sql

Test Data Insertion:
https://github.com/wbt6/docker-database-project/blob/main/init-db/02_insert_data.sql

Technology Components: Software and Hardware Components

Frontend
HTML, CSS, JavaScript

Bootstrap

Backend
PHP (changed due to more experience with PHP)

Database
PostgreSQL

DevOps & Hosting
Docker & Docker Compose

Render (Cloud Hosting)

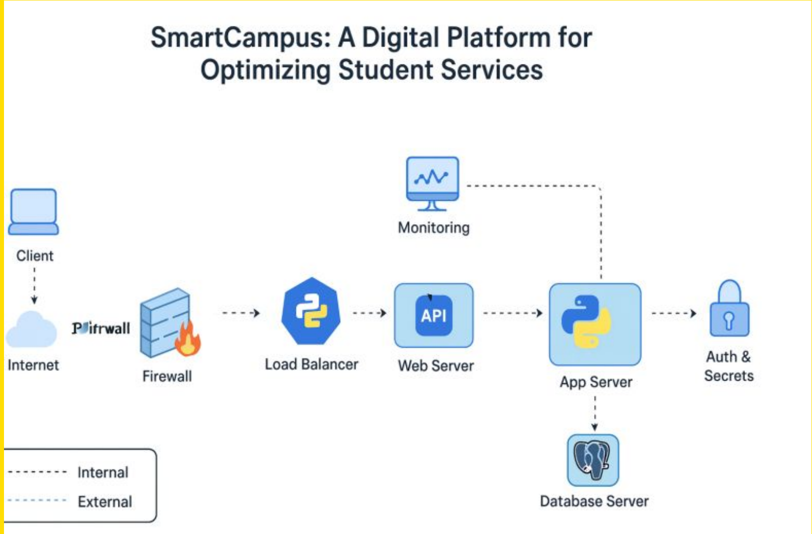
GitHub Actions (CI/CD)

Hardware
Client Devices: Laptop, tablet, mobile

Servers
App Server: 2 vCPU, 4–8 GB RAM

DB Server: 4 vCPU, 16 GB RAM

Architectural Blueprint: System Architecture Diagram



Chosen Architecture
Thin-Client Web + Stateless Application Server (3-Tier)

Architecture Layers
Presentation Layer: Web Browser
Application Layer: API Server
Data Layer: PostgreSQL Database

Why This Architecture?
Scalable (horizontal scaling)
Secure (SSO + encryption)
Maintainable
Supports future cloud expansion (Render)

Trade-Off
More setup than monolithic system
But far more scalable and maintainable long-term

User-Centric Design: User Interface Details

Design Principles

- WCAG 2.1 AA Accessibility
- Mobile-first design
- High-contrast colors
- Large readable fonts

Key Screens

- Student Dashboard
- Booking Page
- Staff Availability Manager
- Ticket Submission Page
- Admin Reports

Design Trade-Offs

- Simple UI over advanced animations
- Faster performance over heavy graphics
- Accessibility prioritized over visual complexity

Github code,project and workflow, Dockerhub image, Live demonstration

Github Project: https://github.com/users/wbt6/projects/2

Code Repository: https://github.com/wbt6/docker-database-project

Dockerhub: https://hub.docker.com/r/wbt6/smartcampus/tags

Live Site: Smart Campus

Prototype:

https://www.figma.com/make/FXrv1C0eyUzTLisjhdRkUO/Ticketing-Feedback-System?node-id=0-1&p=f&fullscreen=1

Q&A

Any questions?